VICTORY 2027

Mineral Resources

LECTURE-2 Geography

Sameer Sir



Topics To Be Covered

1. Copper, Bauxite
2.
3.
4.

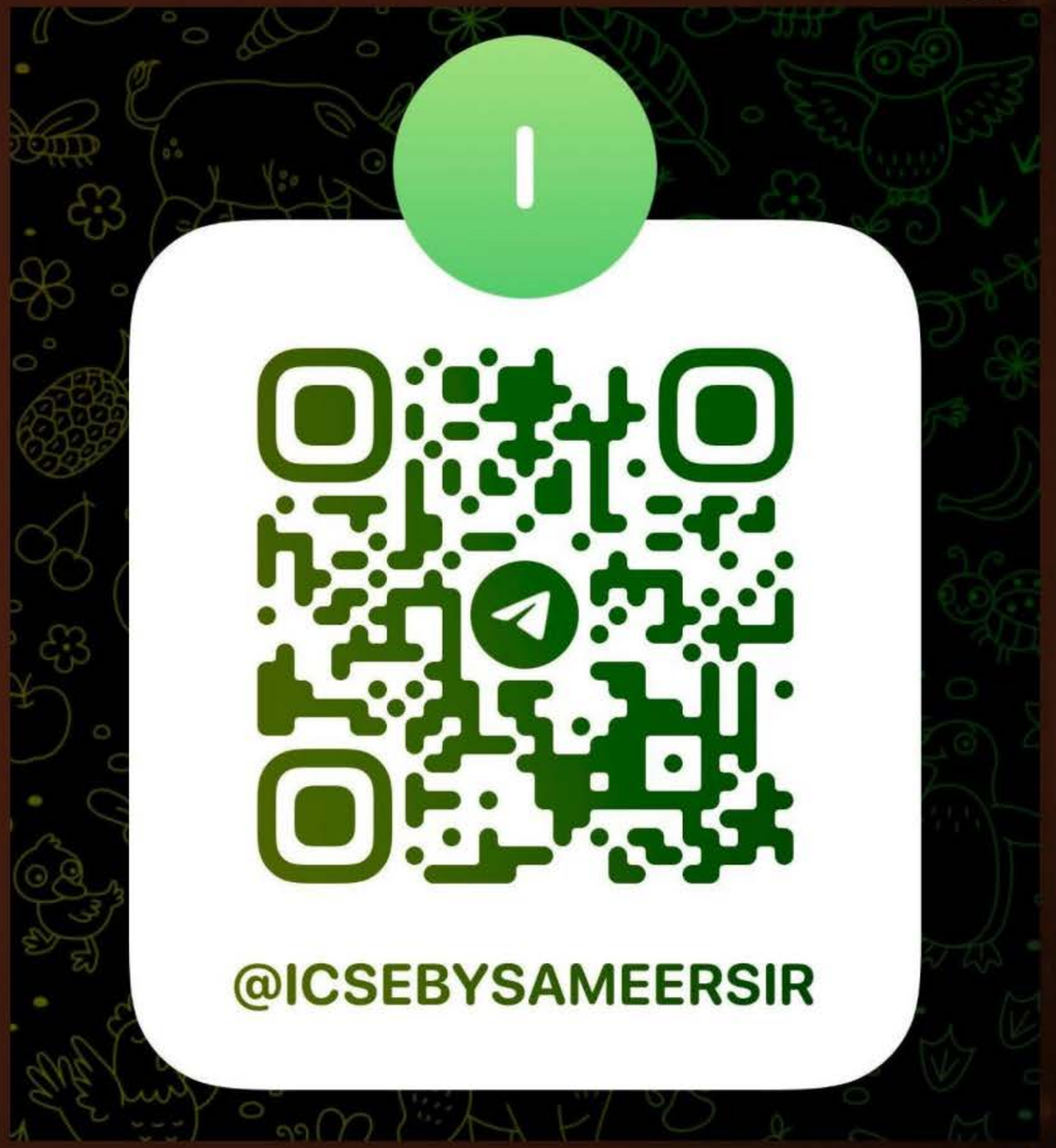
Mineral and Energy Resources

- *Iron ore, Manganese, Copper, Bauxite – uses and their distribution. (two leading states and their prominent mining areas)*
- *Conventional Sources: Coal, Petroleum, Natural gas (distribution, advantages and disadvantages).*
- *Hydel power - Bhakra Nangal Dam and Hirakud Projects and their benefits.*
- *Non-conventional Sources: Solar, wind, tidal, geo-thermal, nuclear and bio-gas (important areas and advantages and disadvantages).*



Telegram

https://t.me/icsebysameersir





Copper

Stone tools

4000 BCE → Copper = Chalcolithic

4600 BCE → Bronze

1000 BCE → Iron

- Copper is an important non-ferrous metal and the earliest metal used by man.
- **In nature, copper occurs in the pure form and in three principal combinations, that is, sulphides, oxides and carbonates.**


India: Copper and Bauxite
Copper
Bauxite
Sikar
Bharatpur
Ajmer
Jaipur
Pali
Bhiwara
Bundi
Chittorgarh
Udaipur
Dungarpur
Tropic of Cancer
Palamau
Jamnagar
Kheda
Jabalpur
Shahdol
Markantak
Mandla
Bilaspur
Singhbhum
Betul
Balaghat
Raigarh
Nagpur
Durgrg
Surat
Bhandara
Sambalpur
Chandrapur
Thane
Kalahandi
ARABIAN SEA
BAY OF BENGAL
Satara
Ratnagiri
Kolhapur
Pangem
Belagavi
Guntur
Mopa
Prakasam
LAKSHADWEEP SEA
LAKSHADWEEP (INDIA)
ANDAMAN AND NICOBAR ISLANDS (INDIA)
ANDAMAN SEA
Salem
Nilgiri
Coimbatore
Madurai
INDIAN OCEAN
SCALE 1:24 000 000
Km 0 240 480 720 Km

Uses

Copper +

① Iron + Nickel → stainless steel

① Copper + Zinc → Brass (तांबा)

① Copper + tin → Bronze (कांस्य)

(i) It is a good conductor of electricity, highly malleable and ductile and is therefore, used for making electric wires and other equipment. ↗ Can be Converted into sheets or wires

(ii) It is used in automobile and defence industries.

(iii) Alloyed with iron & nickel → stainless steel (utensils, objects).

(iv) It is alloyed with zinc to form brass and with tin to make bronze. (तांबा)

(v) It is used in building construction for roofing and plumbing.

(vi) It is used in shipbuilding industry.

Distribution

India is critically deficient in copper and has to depend on exports for its needs. The main areas where copper is found in India are:

State	Important Mining Areas
Madhya Pradesh (LARGEST producer)	Balaghat, Betul, Jabalpur (Malanjkhand mines — Balaghat)
Rajasthan → २	Khetri belt (Jhunjhunu), Dariba (Alwar); also Ajmer, Bharatpur, Bhilwara, Bundi, Chittorgarh, Dungarpur, Jaipur, Pali, Rajsamand, Sikar, Sirohi, Udaipur
Jharkhand	Singhbhum district
Maharashtra	Bhandara, Chandrapur, Gadchiroli, Nagpur
Minor deposits	Gujarat, Karnataka, Andhra Pradesh, Odisha, Uttar Pradesh, Sikkim, Meghalaya, West Bengal

Bauxite → Aluminium oxide



Bauxite

Geology → Study of Earth's Crust, Surface



- Bauxite is an oxide of aluminium. Bauxite is the primary raw ore used to produce aluminium metal.
- Chemically, the aluminium inside it is locked with oxygen (and water molecules), giving it a clay-like or rusty-red appearance rather than a shiny metallic one.
- It is found mainly in tertiary deposits and is associated with laterite rocks occurring extensively either on the plateau or hill ranges of Peninsular India and also in the coastal tracts of the country.

New → 66m - 2.5mya

Laterite rocks → Intense heat + Rain fall!

↓

Laterite Soil

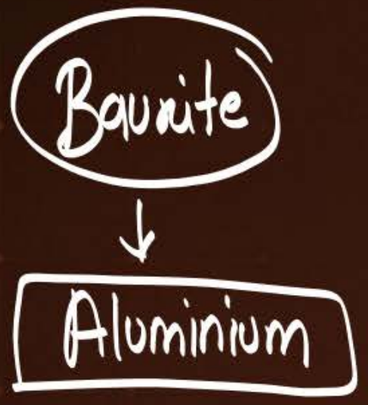
↓

Bricks

Leaching



Uses



(i) Bauxite is the main source of aluminium, which is a lightweight, strong and rust resistant metal.

(ii) Aluminium is used in aircrafts, automobiles, rail wagons, coaches, shipping industry and household appliances.

(iii) It is used in electrical equipment industry and for transmitting electricity because it is a good conductor of electricity.

(iv) Pure aluminium is the best material for making headlight reflectors and mirrors and in telescopes.

Distribution

There are extensive deposits of bauxite in India. The main reserves are found in the following States:

State	Important Areas
Odisha (LARGEST producer)	Kalahandi, Sambalpur
Gujarat	Jamnagar, Kheda, Surat, Kutch
Madhya Pradesh	Jabalpur, Balaghat, Shahdol, Mandla, Amarkantak Plateau
Chhattisgarh	Durg, Bilaspur, Raigarh
Jharkhand	Palamau, Ranchi, Lohardaga, Gumla, Latehar, Dumka
Maharashtra	Thane, Kolhapur, Ratnagiri, Satara
Karnataka	Belagavi (mainly Karle hills)
Tamil Nadu	Salem, Nilgiri, Madurai, Coimbatore
Goa	Mopa, Pernem

Bauxite

The largest integrated aluminium plant in India is at Renukoot in Uttar Pradesh. It gets its supply of bauxite from Amarkantak plateau and Ranchi.

Feature	Iron Ore	Manganese	Copper	Bauxite
Type	Ferrous metallic	Ferrous metallic	Non-ferrous metallic	Non-ferrous metallic
Chemical nature	Oxide of iron	Natural oxide (Mn)	Sulphide/oxide/carbonate; also native	Oxide of aluminium
Best variety	Magnetite (>70% Fe)	—	Native copper (purest)	—
#1 State (current)	Odisha	Odisha	Madhya Pradesh	Odisha
Main use	Steel-making	Toughening steel / ferro-alloys	Electric wires (conductor)	Aluminium extraction
Rock association	Precambrian banded formations, hill masses	Occurs with iron ore belts	Sulphide/oxide veins	Laterite / tertiary rocks, plateaus
India's global status	Among top producers (rich)	Rich, good quality, exported	Deficient — imports needed	Rich, extensive reserves
Key location fact	Exported via Mormugao & Visakhapatnam	Exported to USA, Japan, UK, France	Khetri belt (Rajasthan) famous	Renukoot (UP) — largest Al plant

Bauxite is the ore of:

(a)Aluminium
(b) Copper
(c) Manganese
(d) Iron

A. Iron ore is called versatile mineral. Give reasons.

B. Name the state that is the largest producer of iron ore in India. → Odisha

C. Name any one mining area of Iron ore from the state mentioned by you. Bellur Chikmanglur

Copper → use → ②
↳ Largest producer → 2 mining areas

Thank You